SUSHIL V

Erode, Tamil Nadu
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Career Objective

Motivated Computer Science undergraduate seeking an entry-level software or machine learning role to apply strong problem-solving, programming, and system design skills while gaining real-world industry exposure and contributing to impactful projects.

Education

B.Tech in Computer Science Engineering — Amrita Vishwa Vidyapeetham, Coimbatore	2023 – 2027
Higher Secondary (HSC) — CS Academy, Erode	2023
Secondary (SSLC) — The B.V.B School	2021

Technical Skills

Programming Languages: C, C++, Python, Java, JavaScript, SQL
Web & Application Development: HTML5, CSS3 (Tailwind), Node.js, Electron.js, JWT
Databases: MySQL, MongoDB
IoT & Embedded Systems: ESP32, Arduino, RFID, ThingSpeak
Tools & Platforms: Git, Docker
Machine Learning / CV: CNN, RNN (LSTM), OpenCV, TensorFlow Lite

Projects

CivicTrack – Community Issue Reporting Platform **(May 2025 – Oct 2025)**

– Developed a full-stack MERN application enabling citizens to report and track civic issues.

– Implemented JWT-based authentication and role-based access control.

– Integrated Leaflet.js for interactive geolocation and visualization of reported problems.

– Designed RESTful APIs for CRUD operations, issue lifecycle management, and user handling.

Scene Text Detection Using CNN–RNN **(Apr 2025 – Sept 2025)**

– Built a deep learning pipeline combining CNN feature extraction with LSTM-based sequence modeling.

– Applied OpenCV-based preprocessing for noise reduction and color segmentation.

– Evaluated over 50 diverse video datasets using metrics such as accuracy, CPU usage, memory usage, and latency.

– Achieved approximately 92% character recognition accuracy on test data.

Latency-Aware Task Scheduling for Real-Time Traffic Monitoring (Edge Computing) **(Jun 2025 – Oct 2025)**

– Designed an edge-based anomaly detection system using Raspberry Pi and TensorFlow Lite.

– Reduced detection latency to under 200 ms and network bandwidth usage by over 95%.

– Implemented MQTT-based communication for lightweight alert transmission.

– Enabled real-time event snapshots and centralized dashboard monitoring.

IoT-Based Real-Time Heart Rate Monitoring System **(Jun 2024 – Sept 2024)**

– Programmed ESP32 to capture heart rate and SpO_2 data using a pulse oximeter.

– Integrated ThingSpeak for real-time cloud logging and visualization.

– Designed alert mechanisms for abnormal vitals and generated periodic analytical reports.

RFID-Based Voting System (Embedded Systems) **(Aug 2025 - Nov 2025)**

– Designed and implemented a secure RFID-based electronic voting system using embedded controllers.

– Used RFID cards for voter authentication to prevent duplicate or unauthorized voting.

– Integrated LCD display and control logic to register, validate, and count votes accurately.

– Ensured reliable real-time vote storage and result computation with minimal latency.

Research & Publications

Hybrid CNN–RNN and DCT-Based Framework for Robust Scene Text Detection and Recognition (Under Review)

– Proposed a hybrid deep learning pipeline for improved text recognition in compressed video streams.

– Designed adaptive DCT-based color separation to reduce compression artifacts.

– Improved robustness against quality degradation through region-aware resource allocation.

Certifications

– The Web Developer Bootcamp 2025 (Ongoing)

– Power BI for Beginners

– Learn Ethical Hacking From Scratch (Ongoing)

– Introduction to Cloud Computing on AWS for Beginners — Udemy (Ongoing)

Achievements

– 2^{nd} Place – SIP Regional Prodigy Competition

– 3^{rd} Place – SIP District Prodigy

– Completed all Dakshin Bharath Hindi Prachar Sabha (DBHPS) examinations

Hobbies

Problem Solving • Reading Technology Blogs • Team Sports

Languages

English, Tamil, Malayalam, Hindi